

Term Project Deliverable #4: Repository and Portfolio Submission

Start Assignment

- Due Friday by 11:59pm
- Points 50
- Submitting a website url
- Available until Jun 12 at 11:59pm

Overview

You will use this submission box to upload the URL for your web-hosted repository, which itself includes a URL to the portfolio/documentation for the project.

Portfolio

In addition to the Doxygen-produced documentation in your portfolio, highlight the following things:

- Major Hardware (Electrical *and* Mechanical)
 - Motors or other actuators
 - Sensors
 - Your custom PCB
 - The design of your robot/contraption
- Software Implementation
 - Any drivers that are especially elegant or neatly implemented
 - An overview of your main loop and how you are implementing the intelligence running your project
 - A discussion of your coding style (C++, C, object-oriented, FSMs, tasks, etc.)
- Any mathematical or physical modeling that you have done to interpret sensor data
 - What calculations were performed for dynamic behaviors (e.g., actuation)?
 - Do you have to process sensor data?

The list above should not be considered as an exhaustive checklist. That is, you should include most of the items above plus any other similar items that your team deems appropriate. You should touch on each topic with one or two paragraphs doing your best to concisely but thoroughly explain the topic. Adding in figures will make this part easier.

Repository

Your repository should, at a minimum, include the following:

- All of your source code, ideally in a folder structure compatible with the CubeIDE
- A properly set up .gitignore file
- A properly set up Doxyfile
- A readme file contextualizing your repository along with providing a link to your documentation

Final Notes

- **NO LATE WORK WILL BE ACCEPTED.** Submit early, or risk the alternatives.
- It is probably best for each team mate to have their own portfolio and repository, but only one needs to be submitted and graded for this class.
- As mentioned in lecture, please use Dr. Ridgely's ME405 documentation and repository as a template.
 - [Dr. Ridgely's ME 405 Support Documentation](https://spluttflob.github.io/ME405-Support/) ↗ (https://spluttflob.github.io/ME405-Support/)
 - [Dr. Ridgely's ME 405 Support Repository](https://github.com/spluttflob/ME405-Support) ↗ (https://github.com/spluttflob/ME405-Support)

Term Project Grading Rubric

Criteria	Ratings					Pts
Performance/Functionality Does the robot or project work as designed? Does it perform well? Does it work reliably and repeatedly? Have you diagnosed the problems that occurred? What was the root cause for each issue?	10 to >7.0 pts Excellent Works great, seemingly every time.	7 to >5.0 pts Great Works well, but not every time or works reliably but with lessened performance	5 to >3.0 pts Good Performance is adequate but intermittent, or it works really well but only occasionally.	3 to >0.0 pts Poor Performance is very lacking or there is no reliability in operation.	0 pts Incomplete Project performance is not demonstrated or the demonstrations show serious malfunction.	10 pts
Features/Objectives/Challenges/Workarounds What features did you set out to include? Was the objective ambitious enough for this class? Did the scope of work scale properly with your group's size? If your team found roadblocks or challenges, how effective was your team at overcoming the issues? Did you learn from the issues for the future?	20 to >16.0 pts Excellent Challenges are well presented and strategies to overcome the challenges are well motivated even if they may not completely resolve said challenges.	16 to >10.0 pts Great Challenges are presented, but with reduced detail. Strategies to overcome the challenges are reasonably well motivated but may have short comings or not pursued effectively.	10 to >6.0 pts Good Challenges are vaguely presented or strategies to address challenges are poorly motivated or not mentioned.	6 to >0.0 pts Poor Challenges are not highlighted or are diminished/ignored. Strategies to address challenges are missing or show a lack of motivation to fix the problem.	0 pts Incomplete No challenges have been presented or any challenges presented have no attempted resolution	20 pts
Memo/Documentation Does the memo describe the project well? Does it outline all major hardware and software features? Is it in memo format? Were any interesting or novel strategies depicted (regardless of them being implemented)?	10 to >0.0 pts Full Marks			0 pts No Marks		10 pts
Videos/Attachments Does the project move and behave properly, as shown in a recorded video? Do you have photographs of the completed board? What about renders of the completed CAD design? If you had to do rework, did you take photographs for documentation?	5 to >0.0 pts Full Marks			0 pts No Marks		5 pts
Firmware Is the code organized and well thought out? Does the code illustrate some of the topics covered throughout the course? Does it show good intention and avoid lazy coding practices?	5 to >0.0 pts Full Marks			0 pts No Marks		5 pts
Total Points: 50						